COMP 018: Database Administration - Activity 4 Progress Report
Access Control, Database Users, Roles, Privileges, and Security Governance

Project Identification & Metadata

Course : COMP 018 - Database Administration
Evaluating Group : Group 4: Olivarez, Abangan, Pocsidio, Infantado, and Bartolata
Evaluating Group Repository : https://github.com/Nycri/brewpos
Target Project Monitored : OJT Management System (Project-OJT-RMA-System)
Target Project Group : Group 3 (Evaluated under Tracker Key: Group 06 / Group 03): Simon, De Dios, Ramos, Dela Rosa, Sietereales, Geraldez
Target Project Repository : https://github.com/swvenz/Project-OJT-RMA-System (Branches: main, ojt-system-untested.demo)
Primary Database : ojt_management (PostgreSQL 18.6 on x86_64 Windows, compiled by MSVC)
Restored Test Database : ojt_management_restore_test

ACTIVITY 4: Database Users, Roles, Privileges, and Access Control

1.1 Access-Control Plan & Architecture

Operating an academic internship management and evaluation system under an unrestricted superuser account severely violates the Principle of Least Privilege. In the initial project setup, application configuration files pointed directly to administrative credentials. An attacker or unauthorized student exploiting an unhandled SQL parameter would possess unrestricted data definition and manipulation capabilities across the database cluster, threatening student rosters, logged internship hours, and supervisor evaluations.

Review of the codebase revealed that the application relies entirely on soft-deletion mechanisms using timestamp columns and active flags, never executing physical row deletion statements. Consequently, granting SQL deletion privileges to the application runtime account is unnecessary and introduces severe security risk.

To secure the OJT Management System, a role-based access control architecture was implemented. This plan establishes group roles without login capabilities to manage privilege templates, alongside dedicated login accounts assigned to their respective group roles. The application runtime account receives read, insert, and update permissions across operational tables but is granted zero deletion privileges, providing an absolute guarantee that runtime operations cannot purge student records, attendance logs, or evaluations. Student portal accounts are granted submission and tracking capabilities on specific tables without access to system configurations or administrative credentials. Read-only auditing accounts receive selection rights across operational tables while being denied access to user credential tables to shield password hashes. Logical database backups are assigned to a dedicated read-only

backup account.

1.2 Setup Execution & Privilege Structure

The configuration process established four non-login group roles representing application runtime, student portal, read-only audit, and backup profiles, followed by the creation of dedicated login accounts with secure credentials. Access privileges were revoked from public roles across the database and schema to enforce explicit role grants. Connection and schema usage rights were assigned to designated group roles, explicit table grants were configured to enforce least privilege without deletion rights, and default schema privileges were established to preserve access rules across future database objects.

1.3 Verification Results: Runtime Account Testing

Verification testing demonstrated that the application runtime account successfully queries partner company records, submits new internship applications while advancing sequence counters, and updates application statuses. When row deletion operations were attempted on student profiles, attendance records, or supervisor evaluations, the database engine denied execution under the zero-delete policy. Data definition operations, such as table drop requests, were rejected due to missing object ownership permissions.

1.4 Verification Results: Read-Only Auditor Account Testing

Inspection testing confirmed that the read-only auditing account successfully executes multi-table queries joining student placement details, company information, and evaluation scores. When query attempts were made on user credential tables containing authentication hashes, access was denied by the database security layer. Attempted modification operations, including inserting new requirements, updating evaluation scores, or deleting company records, were uniformly blocked by the engine.

1.5 Verification Results: Cross-Database & Administrative Boundary Testing

Administrative isolation testing showed that database creation commands attempted under application accounts were denied due to missing administrative creation privileges. Connection attempts to unrelated databases on the server instance failed because public connection permissions were revoked, confining access strictly to authorized targets.

1.6 Issues, Fixes, and Feedback to Owners

During initial deployment, an artifact mismatch was resolved where the database file on the main branch contained an unrelated demo lending dump; the official schema definition was retrieved from a secondary project branch and reconciled to fix missing supervisor table columns. Hardcoded superuser credentials in application property files were documented and replaced with least-privilege role credentials.

Feedback provided to Group 3 commended the comprehensive business domain model and soft-deletion architecture. Recommendations advised updating the main repository branch with the official schema DDL script, resolving missing supervisor table columns within database

creation scripts, and implementing dynamic environment property loading rather than bundling hardcoded credentials inside compiled application files.

CLASS GOOGLE SHEETS TRACKER ENTRY (Activity 4)

Column Header	Value
Project	Group 3 - OJT Management System (Project-OJT-RMA-System) [Tracker Key: Group 06 / Group 03]
Repository cloned	https://github.com/swvenz/Project-OJT-RMA-System, Branches: main, ojt-system-untested.demo
Database setup result	Success with issues (Resolved repository artifact mismatch where OJT Database.sql on main was an unrelated demo lending dump; extracted official schema from ojt-system-untested.demo, repaired supervisor column discrepancies, and deployed full 12-table relational schema with RBAC).
Access-control folder link	https://github.com/Nycri/brewpos/tree/main/access-control
Access-control status	Completed
Feedback sent to owners	(05/11/2026)
PDR progress document	Activity4_Progress_Report_Group3.pdf
Last updated	2026-11-05

AI USAGE DECLARATION

Google Gemini 3.8 Flash was utilized during project evaluation to inspect database schema definitions, evaluate relational hierarchy structures, formulate role verification test scenarios, analyze database maintenance operations and query performance execution plans, verify backup creation and restoration procedures, and structure technical report documentation. All database statements, configuration checks, and verification procedures were directly executed and validated against PostgreSQL 18.6 within the local database environment.